

INFO5995 Assignment 1 - Activity Log

Contribution Matrix (N×N Format)

Team Members: Jiakai Bao, Clarissa Mo, Yi Qiao, Jiazhou Shen, Lina He, Handong Ding

N×N Contribution Matrix

Member Jiakai Bao's Assessment

Member	Contribution Description
Jiakai Bao	Task1: Decompiling APK and using AI for preliminary vulnerability analysis Video editing.
Clarissa Mo	Task 2: Application functionality, system architecture, and threat model.
Yi Qiao	Task 3: Detect and analyse vulnerabilities associated with insecure randomness or misuse of cryptographic mechanisms.
Jiazhou Shen	Task 4: Describe the root causes of these vulnerabilities and outline potential attack vectors that could be leveraged by adversaries.
Lina He	Task 5: Propose appropriate mitigation strategies and thoroughly document the identified issues along with the required deliverables.
Handong Ding	Prepare a comprehensive README file and examine all relevant supporting materials.

Member Clarissa Mo's Assessment

Member	Contribution Description
Jiakai Bao	Task 1: APK decompilation, initial vulnerability analysis and presentation video editing.
Clarissa Mo	Task 2: System and threat modeling based on simulator (main) and static analysis.
Yi Qiao	Task 3: Vulnerability analysis related to randomness and cryptographic misuse.
Jiazhou Shen	Task 4: Theoretical analysis on the reason of vulnerability and possible methods of exploitation.
Lina He	

Member	Contribution Description
	Task 5: Mitigation strategies and preparation of submission materials.
Handong Ding	README file and review and validation of submission materials.

Member Yi Qiao's Assessment

Member	Contribution Description
Jiakai Bao	Task 1 APK decompilation (JADX + APKTool setup and execution), reverse engineering of <code>a1_case1.apk</code> , extraction of manifest/package/activity information, location of key classes including <code>MainActivity</code> , <code>Login</code> , and <code>Profile</code> , initial identification of the <code>generateSessionToken()</code> area as a candidate randomness-related weakness, and preparation of the APK analysis ZIP package plus handoff notes for the team.
Clarissa Mo	Task 2 application functionality analysis through emulator interaction and static analysis, including registration/login/profile walkthrough, summary of the app's purpose and main features, identification of core assets such as credentials and session tokens, modelling of data flows between UI, local file storage, and <code>SharedPreferences</code> , and development of the system/threat model focused on local trust boundaries and attacker goals.
Yi Qiao	Task 3 primary vulnerability discovery (CWE-338: <code>java.util.Random</code> for session token), complete vulnerability-chain analysis with 4-step evidence chain (role definition, timing, storage, weak RNG), dynamic PoC validation by collecting real session tokens from <code>SessionPrefs</code> , quantitative attack feasibility analysis (48-bit LCG, 2^{48} seed space, deterministic PRNG reasoning), Task 3 write-up and presentation script drafting, <code>report.pdf</code> contributions (Task 3 section and related system/threat model content), plus repo branch management and merge coordination.
Jiazhou Shen	Task 4 threat model and exploitation analysis — detailed attacker model (rooted-device user / reverse engineer), attacker capabilities and preconditions, explanation of why the token generation method is unsuitable for its security role, construction of a concrete step-by-step attack path from APK reverse engineering to token prediction/forgery risk, and impact assessment covering authentication integrity, impersonation risk, and session hijacking feasibility in principle.
Lina He	Task 5 mitigation strategies and final deliverables support, including proposing the <code>Random</code> → <code>SecureRandom</code> remediation, explaining why the fix works, helping organise the report and presentation materials, documenting the identified issues and recommended fixes, and checking that submission components aligned with assignment requirements and team output.
Handong Ding	README drafting and refinement, supporting-material review, cross-checking submission completeness across Tasks 1–5, validation of PoC logic and feasibility claims, and helping

Member	Contribution Description
	consolidate analysis notes, evidence packaging, and final supporting documents for submission readiness.

Member Jiazhou Shen's Assessment

Member	Contribution Description
Jiakai Bao	Task 1: Worked on decompiling the APK, doing the initial analysis, and editing the presentation video.
Clarissa Mo	Task 2: Worked on analysing the app's functionality and building the system model.
Yi Qiao	Task 3: Worked on identifying the vulnerability related to randomness or cryptographic misuse.
Jiazhou Shen	Task 4: Refined the attacker threat model and explained why the vulnerability exists and how an attacker could potentially exploit it.
Lina He	Task 5: Worked on mitigation, documenting the findings, and preparing the submission materials.
Handong Ding	Prepared the comprehensive README and helped review and consolidate the relevant supporting materials for the final submission.

Member Lina He's Assessment

Member	Contribution Description
Jiakai Bao	Task1 Decompile the APK and initial analyse and presentation viduo editing.
Clarissa Mo	Task2 app's functionality and build system and threat model.
Yi Qiao	Task3 Identify vulnerability related to randomness or cryptographic misuse.
Jiazhou Shen	Task4 Explain why the vulnerability exists and how an attacker could exploit it.
Lina He	Task5 Mitigation and document the findings clearly and required submission materials.
Handong Ding	writing the README and checking and reviewing materials.

Member Handong Ding's Assessment

Member	Contribution Description
Jiakai Bao	Responsible for decompiling the APK using JADX to reconstruct the Java source code. Collaborated with the team to perform preliminary vulnerability scans and managed the final video production.
Clarissa Mo	Defined the application's functional architecture and mapped the data flows between MainActivity, Login, and Profile. Contributed to

Member	Contribution Description
	identifying the protected assets and established the initial threat boundaries.
Yi Qiao	Focused on detecting cryptographic weaknesses, specifically locating the 48-bit PRNG flaw in Login.java and the plaintext storage issue in MainActivity. Collaborated on analyzing the technical root causes.
Jiazhou Shen	Developed the attack vectors and refined the PoC (Proof of Concept) logic to demonstrate how PRNG seed reconstruction leads to session hijacking. Worked with the team to verify the exploitability of identified flaws.
Lina He	Proposed cryptographic mitigation strategies, including the implementation of SecureRandom and increasing token entropy. Responsible for thorough documentation of all identified issues and required deliverables.
Handong Ding	Conducted independent feasibility validation for Tasks 1-5 during the project startup and established the preliminary threat & security models and reporting framework & draft. Collaborated with the team to discuss and refine the logic for the PoC verification methods.

Team Agreement

We confirm that all members contributed roughly equally to this assignment. Each member participated in analysis, documentation, and presentation preparation.